

# Homework 4

Alex Ojemann

2024-10-18

## Problem 1

```
##
## (a) Regression Adjustment with Covariates:

## (a) ATT for Weight change (1971-1982):
## - Estimate: 3.381055
## - Standard Error: 0.4412532
##
## (a) ATT for Systolic Blood Pressure:
## - Estimate: 1.928383
## - Standard Error: 1.034186
##
## (a) ATT for Diastolic Blood Pressure:
## - Estimate: 1.488
## - Standard Error: 0.6202972

##
## (b) Regression Adjustment with Propensity Score:

## (b) ATT for Weight change (1971-1982):
## - Estimate: 3.446452
## - Standard Error: 0.454662
##
## (b) ATT for Systolic Blood Pressure:
## - Estimate: 1.771341
## - Standard Error: 1.115427
##
## (b) ATT for Diastolic Blood Pressure:
## - Estimate: 1.468171
## - Standard Error: 0.6387768

##
## (c) Propensity Score Stratification:

## (c) ATT for Weight change (1971-1982):
## - Estimate: 3.330245
##
## (c) ATT for Systolic Blood Pressure:
## - Estimate: 1.768685
```

```

##
## (c) ATT for Diastolic Blood Pressure:
## - Estimate: 1.424823

##
## (d) Inverse Probability Weighting (IPW):

## (d) ATT for Weight change (1971-1982):
## - Estimate: 3.365535
## - Standard Error: 0.5064518
##
## (d) ATT for Systolic Blood Pressure:
## - Estimate: 1.993023
## - Standard Error: 1.183082
##
## (d) ATT for Diastolic Blood Pressure:
## - Estimate: 1.55159
## - Standard Error: 0.6514146

##
## (e) Propensity Score Matching (1:1):

## (e) 1:1 Matching ATT for Weight change (1971-1982):
## - Estimate: 3.907661
## - Standard Error: 0.5501104
##
## (e) 1:1 Matching ATT for Systolic Blood Pressure:
## - Estimate: 2.188345
## - Standard Error: 1.39249
##
## (e) 1:1 Matching ATT for Diastolic Blood Pressure:
## - Estimate: 1.792001
## - Standard Error: 0.7430079

##
## (e) Propensity Score Matching (2:1):

## (e) 2:1 Matching ATT for Weight change (1971-1982):
## - Estimate: 3.086314
## - Standard Error: 0.4742164
##
## (e) 2:1 Matching ATT for Systolic Blood Pressure:
## - Estimate: 2.598548
## - Standard Error: 1.170929
##
## (e) 2:1 Matching ATT for Diastolic Blood Pressure:
## - Estimate: 1.690736
## - Standard Error: 0.636296

```

## Problem 2

